

Network infrastructure project

Project Title: Advanced Healthcare Enterprise Network Architecture for Hospital Management



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SLIDE 01

PROJECT TEAM NETWORK ARCHITECTURE

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Hospital Enterprise Network

SYSTEM DOMAIN FOCUS

Zero-Trust Medical Data Security

High-Availability Core Backbone

Healthcare Operations

Active

1. Project Overview

This project involves designing and simulating a secure, high-availability network infrastructure for a modern healthcare facility using **Cisco Packet Tracer**. It seamlessly interconnects vital hospital departments to ensure reliable data transmission and patient care continuity.



Department Interconnection

Unifies **Emergency**, **Administration**, **Clinics**, and **Laboratories** into a cohesive communication backbone.

✓ Reliable & Low Latency



VLAN Segmentation

Utilizes Virtual LANs to isolate broadcast domains, preventing network congestion and unauthorized inter-department traffic.

 Isolated Subnets



Patient Data Security

Employs robust security controls like **Port Security** and **SSH Encryption** to shield sensitive medical records.

 Encrypted & Hardened



Cisco Packet Tracer Simulation



Zero-Trust Port Access Security



Multi-Layer Switched Topology

Classful IP Addressing Plan

Structured IPv4 allocation for department VLANs and core link networks

Network / VLAN	Available Hosts	Network Address	Prefix	Usable Range	Broadcast Address
VLAN 10	254 Hosts	192.168.10.0	/24	192.168.10.1 – 192.168.10.254	192.168.10.255
VLAN 20	254 Hosts	192.168.20.0	/24	192.168.20.1 – 192.168.20.254	192.168.20.255
VLAN 40	254 Hosts	192.168.40.0	/24	192.168.40.1 – 192.168.40.254	192.168.40.255
VLAN 50	254 Hosts	192.168.50.0	/24	192.168.50.1 – 192.168.50.254	192.168.50.255
Link Network 1	254 Hosts	192.168.80.0	/24	192.168.80.1 – 192.168.80.254	192.168.80.255
Link Network 2	254 Hosts	192.168.90.0	/24	192.168.90.1 – 192.168.90.254	192.168.90.255
Link Network 3	254 Hosts	192.168.100.0	/24	192.168.100.1 – 192.168.100.254	192.168.100.255
Link Network 4	254 Hosts	192.168.60.0	/24	192.168.60.1 – 192.168.60.254	192.168.60.255
Link Network 5	254 Hosts	192.168.70.0	/24	192.168.70.1 – 192.168.70.254	192.168.70.255



Routing Design: OSPF Area 0 Backbone

The **OSPF dynamic routing protocol** is implemented across all core and distribution switches within the hospital network to ensure automatic route updates, high availability, and efficient inter-VLAN communication.

MULTIL SW -ACTIVE

Active

```
en
conf t
int g1/0/4
no sw
ip add 192.168.60.1 255.255.255.0
no sh
int g1/0/3
no sw
ip add 192.168.80.1 255.255.255.0
no sh
exit
router ospf 1
network 192.168.60.0 0.0.0.255 area
0
network 192.168.80.0 0.0.0.255 area
0
network 192.168.10.0 0.0.0.255 area
0
network 192.168.20.0 0.0.0.255 area
0
network 192.168.50.0 0.0.0.255 area
0
network 192.168.40.0 0.0.0.255 area
0
```

MULTIL SW -STANDBY

Standby

```
en
conf t
int g1/0/4
no sw
ip add 192.168.90.1 255.255.255.0
no sh
int g1/0/3
no sw
ip add 192.168.70.1 255.255.255.0
no sh
exit
router ospf 1
network 192.168.90.0 0.0.0.255 area
0
network 192.168.70.0 0.0.0.255 area
0
network 192.168.10.0 0.0.0.255 area
0
network 192.168.20.0 0.0.0.255 area
0
network 192.168.50.0 0.0.0.255 area
0
network 192.168.40.0 0.0.0.255 area
0
```

D-SW₁

Dist-1

```
en
conf t
int g1/0/1
no sw
ip add 192.168.60.2 255.255.255.0
no sh
int g1/0/2
no sw
ip add 192.168.70.2 255.255.255.0
no sh
exit
ip routing
router ospf 1
network 192.168.60.0 0.0.0.255 area
0
network 192.168.70.0 0.0.0.255 area
0
```

D-SW₂

Dist-2

```
en
conf t
int g1/0/1
no sw
ip add 192.168.80.2 255.255.255.0
no sh
int g1/0/2
no sw
ip add 192.168.90.2 255.255.255.0
no sh
exit
ip routing
router ospf 1
network 192.168.80.0 0.0.0.255 area
0
network 192.168.90.0 0.0.0.255 area
0
```

Security Implementation Matrix

Multi-layered defensive policies protecting switches, routers, and server infrastructure

SECURITY POLICY	TARGET DEVICES	PURPOSE & PROTECTION GOAL	VERIFICATION COMMAND
BPDU Guard + PortFast	Access Switches	Blocks unauthorized switch connection to end-device ports and mitigates STP-based attacks.	show spanning-tree interface detail
STP Root Primary/Secondary	Active & Standby	Explicitly defines root bridge hierarchy, preventing a rogue switch from becoming root.	show spanning-tree summary
Extended ACL (PROTECT-SERVERS)	Active & Standby	Restricts access to critical servers (DHCP, DNS, HTTP, FTP) to authorized protocols/ports.	show access-lists PROTECT-SERVERS
Anti-Spoofing ACL	Router2 & Router3	Blocks inbound traffic carrying private/spoofed source IP addresses from entering.	show access-lists ANTI-SPOOF-IN
SSH Configuration	All Devices	Provides encrypted remote management access to devices, replacing plaintext Telnet.	show ip ssh / show ssh
Enable Secret & Passwords	All Devices	Encrypts privileged EXEC mode and secures console/VTY lines against unauthorized logins.	show running-config
Port Security	Access Switches	Restricts switch ports to authorized MAC addresses only, preventing MAC flooding.	show port-security interface
Disabling Unused Interfaces	Access Switches	Enhances physical security by administratively shutting down all unused ports (shutdown).	show ip interface brief

Network Services: DHCP & DNS Integration

Centralized IP dynamic configuration and domain name resolution for all hospital workstations and servers.

Automated DHCP Service

A central **DHCP Server** automatically assigns IP addresses, subnet masks, default gateways, and DNS settings to all client devices across hospital departments, eliminating manual IP conflict errors.

Domain Name Service (DNS)

A dedicated **DNS Service** maps medical server hostnames to their respective IP addresses, enabling healthcare personnel to access medical databases via user-friendly domain names.

PC1 - IP Configuration

DHCP SUCCESSFUL

IP Assignment:	DHCP Dynamic
IPv4 Address:	192.168.20.22
Subnet Mask:	255.255.255.0
Default Gateway:	192.168.20.254
DNS Server:	192.168.100.2

Service Verification Status

End-user client PC successfully acquired IP configuration and resolved hospital server hostnames without packet loss.

7. Connectivity Verification & Ping Testing

Comprehensive ICMP echo request testing confirming full reachability across VLANs, core switches, and central server subnets.

 **Inter-VLAN Routing Test**
Ping from PC1 (VLAN 20) to PC2 (VLAN 10)

✓ 100% Success

 **Server Farm Reachability**
Ping to DHCP (192.168.100.2) & Web (192.168.100.4)

✓ 100% Success

 **Core Switch Links**
Ping across 192.168.60.0/24 & 192.168.80.0/24

✓ 100% Success

 **Default Gateway Ping**
Ping to HSRP Virtual IP 192.168.20.254

✓ 100% Success

PC1 Command Prompt - ICMP Diagnostics

ACTIVE

Packet Tracer PC Command Line v7.3

PC1> ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=<1ms TTL=128
Ping statistics for 192.168.10.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

PC1> ping 192.168.100.4

Pinging 192.168.100.4 with 32 bytes of data:
Reply from 192.168.100.4: bytes=32 time=2ms TTL=127
Ping statistics for 192.168.100.4: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

✓ All Subnets Reachable

Average RTT: 1.2ms



8. Web (HTTP) & FTP Server Access Verification

Validating end-user access to central hospital web portal and secure FTP file transfer repository.

The image is a screenshot of a web browser displaying the 'Healthcare Enterprise Portal'. The address bar at the top shows the URL 'http://www.hospital.com' with a green padlock icon on the left and '200 OK' in a green box on the right. The main content area has a dark blue header. In the center of the header is a white icon representing a medical professional (a person silhouette next to a screen with a plus sign). Below the icon, the text 'Healthcare Enterprise Portal' is written in a large, bold, white font. Underneath that, in a smaller white font, is 'Welcome to Central Medical Information System'. Below the header, there are two light purple status bars. The first bar contains the text 'Server Host:' followed by '192.168.100.4' in a purple font. The second bar contains the text 'Service Status:' followed by 'HTTP Port 80 Active' in a green font. At the bottom of the browser window, there is a purple bar with a white checkmark icon and the text 'Web Portal Operational' on the left, and 'DNS Resolved' on the right.

FTP File Transfer Session (Port 21)

AUTHENTICATED

```
PC1> ftp 192.168.100.3
Connected to 192.168.100.3.
220 Cisco FTP Server Ready.
User (192.168.100.3:(none)): admin
331 Password required for admin.
230 User logged in.
ftp> dir
patient_records.db      4096 KB
lab_results_v2.xml      1024 KB
ftp> put report_2026.pdf
226 Transfer complete. 512 bytes sent in 0.01 secs.
```

 Secure File Exchange Verified

Read / Write Granted

9. HSRP High Availability Configuration

Deploying Hot Standby Router Protocol to eliminate single points of failure across default gateways.



MULTIL SW - ACTIVE

Priority 110 (Preempt Enabled)

STATE: ACTIVE



MULTIL SW - STANDBY

Priority 100 (Default)

STATE: STANDBY



Virtual Gateway IP

192.168.x.254

GROUP 10/20

>_ Active Switch CLI (VLAN 10 HSRP)

Primary

```
MULTIL_SW_ACTIVE(config)# interface Vlan 10
MULTIL_SW_ACTIVE(config-if)# ip address 192.168.10.1 255.255.255.0
MULTIL_SW_ACTIVE(config-if)# standby 10 ip 192.168.10.254
MULTIL_SW_ACTIVE(config-if)# standby 10 priority 110
MULTIL_SW_ACTIVE(config-if)# standby 10 preempt
%HSRP-6-STATECHANGE: Vlan10 Group 10 state Standby -> Active
```

>_ Standby Switch CLI Verification

Backup

```
MULTIL_SW_STANDBY# show standby brief
Interface Grp Pri P State Active Standby Virtual IP
Vlan10 10 100 Standby 192.168.10.1 local 192.168.10.254
Vlan20 20 100 Standby 192.168.20.1 local 192.168.20.254
HSRP State: Active/Standby Pair Fully Converged
```

10. EtherChannel (Link Aggregation) Setup

Bundling physical FastEthernet and Gigabit interfaces into logical Port-Channels for bandwidth aggregation and link redundancy.

AGGREGATED SPEED

2 Gbps Trunk

CONTROL PROTOCOL

LACP (802.3ad)

PORT-CHANNEL STATUS

Po1 (SU) In Use

</> Core Switch EtherChannel ConfigLACP Active

```
Core-SW1(config)# interface range g1/0/1 - 2
Core-SW1(config-if-range)# channel-group 1 mode active
Core-SW1(config-if-range)# exit
Core-SW1(config)# interface port-channel 1
Core-SW1(config-if)# switchport mode trunk
Core-SW1(config-if)# switchport trunk allowed vlan 10,20,40,50
%LINK-3-UPDOWN: Interface Port-channel1, changed state to up
```

Verification: show etherchannel summaryActive Links

```
Core-SW1# show etherchannel summary
Flags: D - down      P - in port-channel    I - stand-alone
        H - Hot-standby  s - suspended    SU - Layer2/In-use

Group  Port-Channel  Protocol  Ports
1      Po1(SU)      LACP      Gig1/0/1(P) Gig1/0/2(P)

Status: Both physical interfaces operational and bundled.
```

11. Wireless Network (WLAN) Configuration

Extending hospital infrastructure mobility to medical handhelds, carts, staff laptops, and guest users.



Medical Staff WLAN

SSID: Hospital-Staff

Mapped to **VLAN 10**. Secured with WPA2-PSK encryption. Provides doctors and nurses direct access to electronic medical records (EMR).

Subnet: 192.168.10.0/24

Secure



Medical IoT & Telemetry

SSID: Hospital-IoT

Mapped to **VLAN 40**. Dedicated high-priority wireless channel for mobile patient monitoring telemetry and vital sign sensors.

Subnet: 192.168.40.0/24

Isolated



Guest Wi-Fi Access

SSID: Hospital-Guest

Mapped to **VLAN 50**. Restricted by extended ACLs to prevent access to internal servers and patient database networks.

Subnet: 192.168.50.0/24

Restricted

12. WAN & Internet Edge Configuration

Connecting internal hospital subnets to external Internet services using Port Address Translation (PAT).

>_ Edge Router - NAT Overload (PAT) Config

Edge Router

```
Router_Edge(config)# access-list 1 permit 192.168.0.0 0.0.255.255
Router_Edge(config)# ip nat inside source list 1 interface g0/0/0 overload
Router_Edge(config)# interface g0/0/1
Router_Edge(config-if)# ip nat inside
Router_Edge(config)# interface g0/0/0
Router_Edge(config-if)# ip nat outside
Router_Edge(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

☰ NAT Translation Table Verification

Translations Active

```
Router_Edge# show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp 203.0.113.2:1025    192.168.10.15:1025 198.51.100.8:80    198.51.100.8:80
tcp 203.0.113.2:1026    192.168.20.22:1026 198.51.100.8:443   198.51.100.8:443

Total active translations: 2 (Dynamic PAT Enabled)
Private IPv4 subnets successfully mapped to public ISP address.
```

13. Conclusion & Future Recommendations

Synthesizing network design achievements and charting the next phase of enterprise infrastructure evolution.

Key Design Achievements

- ✔ Zero Downtime Architecture: HSRP failover & EtherChannel trunks guarantee high availability.
- ✔ Hardened Security: Port Security, SSH, BPDU Guard, and ACLs protect patient data.
- ✔ Automated Operations: Integrated DHCP & DNS streamline medical device connectivity.

✔ All Project Objectives Successfully Realized

Future Enhancement Roadmap

- 💡 IPv6 Dual-Stack Migration: Expand address space for high-density medical IoT devices.
- 💡 Next-Gen Firewall (NGFW): Deploy deep packet inspection & intrusion prevention (IPS).
- 💡 SD-WAN Interconnection: Securely connect remote clinic branches to core hospital databases.

🚀 Scalable for Next-Generation Healthcare Infrastructure

Image Sources



https://hga.com/wp-content/uploads/2022/10/4636_001_00_220622CG_HGA_Zotec_010v3_medium-1.jpg

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